Acid–Base Properties of Salts

Answer key with worked solutions

A. Predicting

1. Acidic, basic or neutral.

> (a) **neutral** (no ion hydrolyses) (b) **basic** (CO_3^{2-}) (c) **acidic** (NH_4^+) (d) **basic** (CH_3COO^-)
> (e) **neutral** (f) **acidic** (NH_4^+)

2. Hydrolysis equations.

> (a) $F^-(aq) + H_2O(l) \rightleftharpoons HF(aq) + OH^-(aq)$
> (b) $NH_4^+(aq) + H_2O(l) \rightleftharpoons NH_3(aq) + H_3O^+(aq)$

B. Ka and Kb

3. Ka and Kb.

> (a) $K_b(F^-) = \dfrac{1.0 \times 10^{-14}}{6.8 \times 10^{-4}} = \mathbf{1.5 \times 10^{-11}}$
> (b) $K_a(NH_4^+) = \dfrac{1.0 \times 10^{-14}}{1.8 \times 10^{-5}} = \mathbf{5.6 \times 10^{-10}}$
> (c) $K_b(CN^-) = \dfrac{1.0 \times 10^{-14}}{4.9 \times 10^{-10}} = \mathbf{2.0 \times 10^{-5}}$
> **CN^-** is the stronger base (larger Kb), because HCN is a much weaker acid than HF.

C. pH of salt solutions

4. 0.10 mol/L NaCN.

> $[OH^-] \approx \sqrt{2.0 \times 10^{-5} \times 0.10} = 1.41 \times 10^{-3}$ mol/L (1.4 % of 0.10 mol/L: approximation OK)
> $pOH = -\log(1.41 \times 10^{-3}) = 2.85$; $pH = 14.00 - 2.85 = \mathbf{11.15}$

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5. 0.050 mol/L NH_4NO_3.

> $[H_3O^+] \approx \sqrt{5.6 \times 10^{-10} \times 0.050} = 5.29 \times 10^{-6}\ \text{mol/L}$
> $\text{pH} = -\log(5.29 \times 10^{-6}) = \mathbf{5.28}$

6. 0.25 mol/L CH_3COOK.

> $K_b(CH_3COO^-) = 5.6 \times 10^{-10}$; $[OH^-] \approx \sqrt{5.6 \times 10^{-10} \times 0.25} = 1.18 \times 10^{-5}\ \text{mol/L}$
> $\text{pOH} = 4.93$; $\text{pH} = 14.00 - 4.93 = \mathbf{9.07}$

D. Titrations

7. Equivalence point.

> (a) Same concentration, 1 : 1 reaction: **25.00 mL** of NaOH.
>
> (b) Moles of ethanoate $= 0.100\ \text{mol/L} \times 0.02500\ \text{L} = 2.50 \times 10^{-3}$ mol, in 0.05000 L: $c =$ **0.0500 mol/L**
>
> (c) $[OH^-] \approx \sqrt{5.6 \times 10^{-10} \times 0.0500} = 5.29 \times 10^{-6}$ mol/L; pOH = 5.28; pH = **8.72**. **Phenolphthalein** (changes colour at about pH 8.2–10) is suitable.

E. Challenge

8. Aluminium chloride.

> In water, Al^{3+} is surrounded by six water molecules, $[Al(H_2O)_6]^{3+}$. The small, highly charged ion pulls electron density from the O–H bonds of these water molecules, so one of them can release a proton: $[Al(H_2O)_6]^{3+} + H_2O \rightleftharpoons [Al(H_2O)_5(OH)]^{2+} + H_3O^+$. The H_3O^+ formed makes the solution **acidic**.

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